(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

☘️

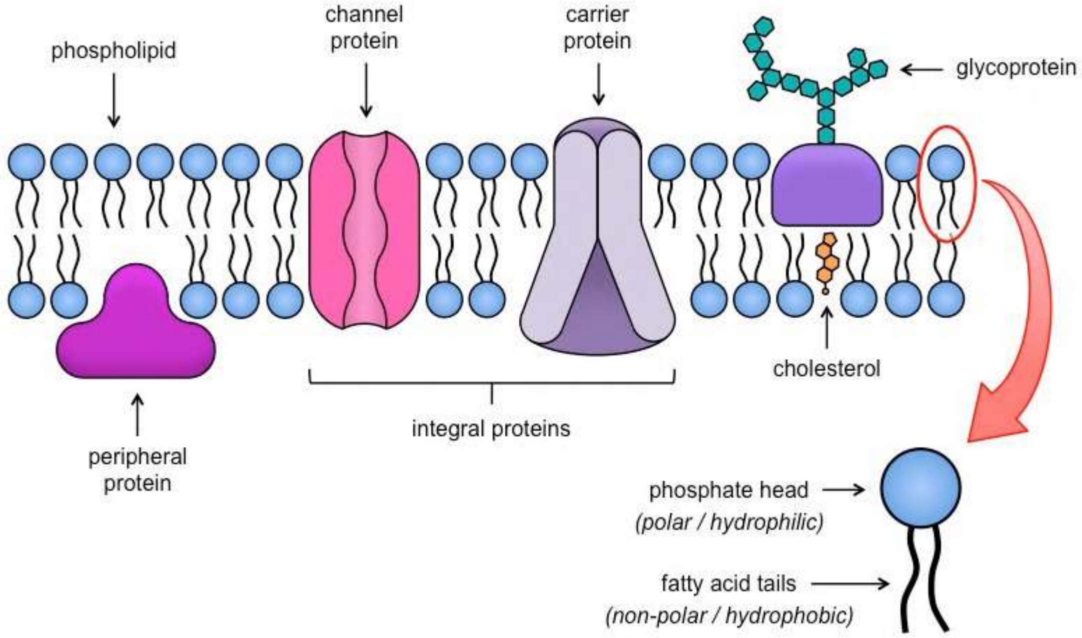
Cell Membrane and Transport

The Fluid Mosaic Model

▼ What is the cell membrane and its funtions?

▼ Explain the Fluid Mosaic Model

The fluid mosaic model is made up of a phospholipid bilayer with phospholipids flowing freely in the plasma membrane and different molecules such as proteins, carbohydrates, cholesterol, etc., randomly embedded or scattered in the membrane.



- Made up of a bilayer/two rows of phospholipids.
- Hydrophillic phosphate group head faces outside/extracellular fluid and attracts water.
- Hydrophobic fatty acid chains face the inside/intracellular fluid and repels water.
- The hydrophillic head and hydrophic tail causes the cell membrane to be selectively permeable and only allow small and non-polar molecules

i.e. O_2, CO_2, etc. to passively pass through.

- It has **cholesterol** which gives the cell membrane stability, makes it flexible and fluid. It also links phospholipids in the bilayer together.
- It contains 2 kinds of proteins:
  1. Peripheral Proteins:
     - Found on the surface of the membrane
     - Helps in cell recognition
     - Act as binding sites for molecules.
  2. Integral Proteins
     - Carrier Proteins: Carries large molecules across the membrane in facilitated or active transport by binding to the molecule, changing its shape and carrying the molecule across the membrane. I.e. for glucose
     - Channel Proteins: Forms a functional pore that acts as a water-filled channel used to carry lipid insoluble, polar molecules i.e. ions in facilitated diffusion.
- It also contains carbohydrates in the form of glycolipids and glycoproteins. Glycoproteins are on the surface of the membrane have carbohydrate chains which help in cell recognition.

▼ Summarise the functions of plasma membrane components

Summary of functions of plasma membrane components

Membrane components	Functions
Phospholipids	• Affect the fluidity and permeability of the membrane. • Barrier to most water-soluble substances • Presence of double bond in tails prevent tight packing
Cholesterol	• Makes the membrane less fluid at higher temperatures but more fluid at lower ones.
Proteins	• Provide structural support for the membrane. • Assist the active transport of materials across the membrane. • Act as recognition sites. (body's immune system) • Act as enzymes and electron carriers. • Used in cell-to-cell adhesion
Glycolipids	• Act as recognition sites. (ABO blood group system) • Help to make the membrane more stable.
Glycoproteins	• Act as recognition sites. (Eg: neurotransmitters and hormones)

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▼ Why is cholesterol important in membrane fluidity?**

Cholesterol has different effects on membrane fluidity at different temperatures.

At warm moderate temperatures (such as 37°C), cholesterol restrains movement of phospholipids by increasing the packing of phospholipids. Cholesterol can fit into spaces between phospholipids and prevent water-soluble molecules from diffusing across the membrane. At cool temperatures, it maintains fluidity by preventing tight packing (solidification)

▼ How does cell recognition work?**

Cells recognize each other by binding to surface molecules, often carbohydrates, on the plasma membrane. Carbohydrates also form hydrogen bonds with water molecules to stabilize the membrane structure.

Glycoproteins act as receptor molecules for substances such as hormones & neurotransmitters.

Glycolipids cause adhesion to neighbouring cells to form tissues and their main role is in stability and to facilitate cellular recognition.

Membrane Transport

▼ Define Passive Transport

Passive transport is the movement of substances from high concentrations to low concentrations without the use of energy/ATP.

3 Types:

1. Diffusion
    - The passive movement of molecules from a region of high concentration to a region of low concentration, down a concentration gradient, through a semi-permeable membrane.
    - In simple diffusion, small, non-polar molecules pass directly through the phospholipid bilayer
2. Osmosis
    - The passive movement of water molecules from a region of high water potential to a region of lower water potential, down a water potential gradient, through a semi-permeable membrane
3. Facilitated Diffusion
    - The passive movement of large, polar molecules from a region of high conc. to low conc. using transport proteins like carrier and channel proteins.

▼ Differentiate between the 3 water potentials

Water potential is the tendency of water molecules to move from one area to another. The 3 Water Potentials are:

1. Isotonic - Solute Concentration inside the cell equals that outside the cell so no osmosis occurs.
2. Hypertonic - Higher solute concentration
3. Hypotonic - Lower solute concentration

▼ What does the rate of diffusion depend on?

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The rate of diffusion depends on:

Surface area of membrane	Explanation
Type of medium	The rate of diffusion is higher in gas than in liquid.
Surface area of membrane	The greater the surface area of a membrane, the faster the diffusion rate
Distance over which diffusion takes place	The greater the distance for a diffusion to take place, the slower the rate of diffusion
Size and nature of the molecules	Larger molecules diffuse more slowly because they require more energy to move around than smaller molecules
Concentration gradient	The larger the concentration gradient, the higher the diffusion rate.
Temperature	At high temperatures, molecules have more kinetic energy. They move around faster, increasing the speed of diffusion

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▼ Compare & contrast both types of diffusion.

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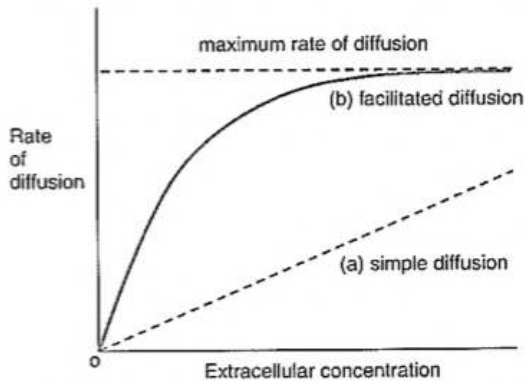
Differences and similarities between simple diffusion and facilitated diffusion



	Simple diffusion	Facilitated diffusion
Similarities	Passive- does not involve energy	Passive- does not involve energy
Differences	**No transport proteins**	**Involves transport proteins**
	Lower rate of diffusion in a high concentrated solution	**Higher rate** of diffusion in a high concentrated solution

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▼ Describe Active Transport

Active Transport is the the movement of molecules from low concentration to higher concentration, against a gradient, through a partially permeable membrane using ATP and carrier proteins.

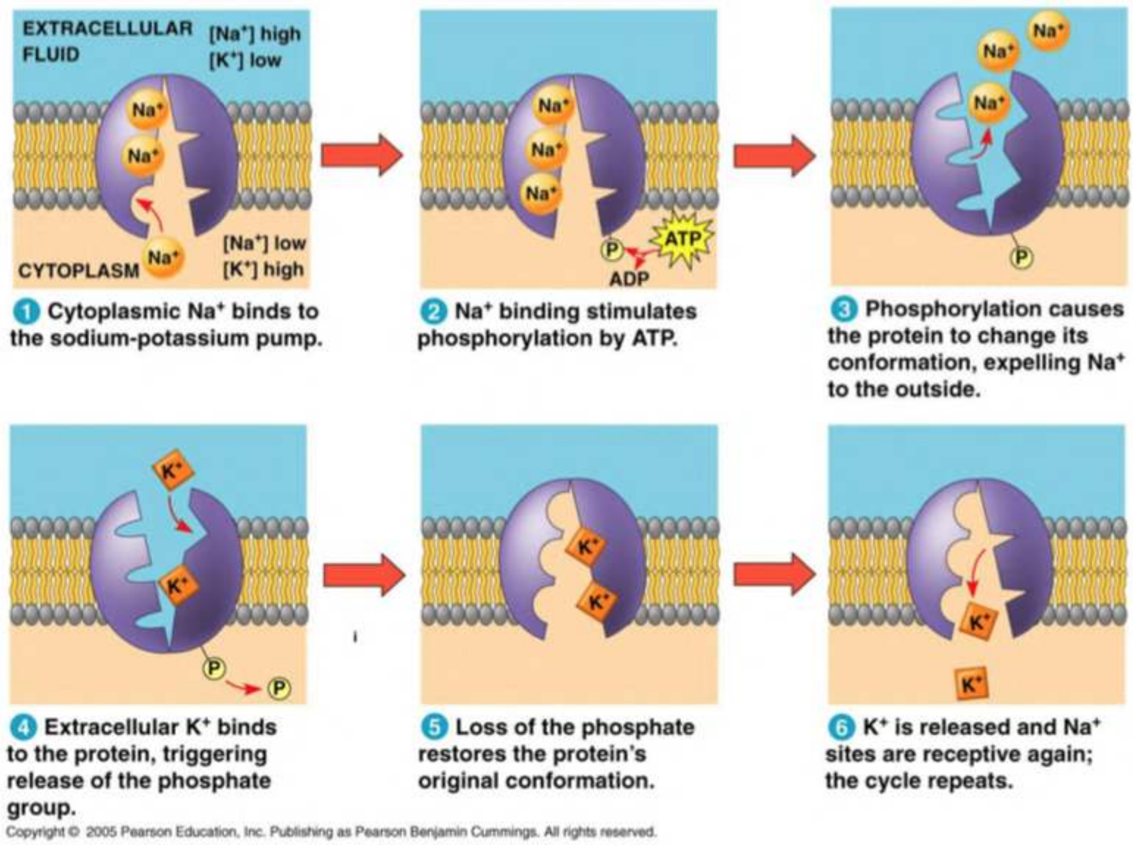
▼ Explain the sodium potassium pump briefly with reference to carrier proteins & ATP.

- Carrier Proteins known as pumps transport substances across the plasma membrane against the concentration gradient using ATP.
- Carrier proteins are specific and have a receptor site specific to the substance it binds to. ATP changes the shape of the protein and causes the substance to be release on the other side.

The Na-K Pump

- Helps in osmoregulation and maintains electrical activity in nerve and muscle cells.
- For every 2 K ions, 3 Na is removed. After nerve impulses are passed, the ion concentrations have to return to normal and this is reset using the Na-K pump and ATP.

1. 3 Na+ ions in the cytoplasm bind to the sodium-potassium pump
2. The sodium binding stimulates phosphorylation by ATP which gives a phosphate molecule to the pump and causes the carrier protein to change its shape.
3. The change in shape causes Na+ to be expelled to the extracellular fluid.
4. 2 Extracellular K+ ions then bind to the protein, which then causes the carrier protein to release the phosphate group.
5. Dephosphorylation causes the protein to revert back to its original shape and causes K+ to be released into the cytoplasm.
6. The cycle repeats as Na+ ions bind again to the carrier protein.



- Define Cytosis and its types

  Cytosis - A form of bulk transport where large molecules that are too big to passively diffuse across the membrane are transported through vesicles which move into and out of cells.

  Endocytosis - Bulk transport of large molecules into the cell using vesicles

  Exocytosis - Bulk transport of large molecules out of the cell using vesicle

- Differentiate between endocytosis and exocytosis

  Endocytosis

  - Moves substances into the cell
  - The cell membrane folds inwards causing a depression to form where large molecules can move into the cell membrane. The membrane then engulfs the material, forming a vesicle which then carries the material into the cell
  - Summary: Molecules are enclosed in vesicles made from the cell surface membrane and transported into the cell.
  - Phagocytosis - Engulfing of solid material

- Pinocytosis - Engulfing of liquid/dissolved substances

Exocytosis

- Moves substances out of the cell
- **Vesicles produced by golgi bodies** containing intracellular materials move outwards and **fuse** with the plasma membrane, allowing the molecules to **leave** from the cell.

▼ Gas Exchange surfaces

The human lung has a large number of alveoli which provide a large surface area for gas exchange to occur through diffusion. The walls of the alveoli are thin and are made up of a single layer of flattened endothelial cells that ensures a short diffusion distance. There is also an extensive network of capillaries that maintain the steep concentration gradient for gas exchange to occur.

▼ Common Past Paper Questions

- The structure and properties of the cell membrane control which molecules can
  move into or out of the cell.

  (a) The phospholipid bilayer plays an important role in this control of movement of
  molecules.

  Explain why the phospholipid molecules form a bilayer.

Question Number	Answer	Additional Guidance	Mark
3(a)	1. {phosphate group / heads} are hydrophilic ; 2. Idea that heads can be attracted to water ; 3. {fatty acids / tails} are hydrophobic ; 4. Idea that tails orientate themselves away from water / eq ; 5. Idea of aqueous environment on both sides of the membrane ;	ACCEPT marks for annotated diagram, phonetic spelling OK IGNORE "water loving / hating" 1. CCEPT polar 2. t just facing water 3. CCEPT non polar 4. CCEPT repel water, face away from water, away from polar environment 5. CCEPT polar environment	(3)

- Describe the structure of a cell membrane.

Question Number	Answer	Additional Guidance	Mark
4(a)		Read what is written on the lines first Accept points made on a clearly labelled diagram If diagram and description contradict then Mp not awarded	
	1. reference to phospholipid bilayer ;		
	2. correct orientation and structure of the phospholipids in the bilayer ;	2. CCEPT heads on outside and each with two tails if drawn 2. N if gap between phospholipids is too large e.g. as large as a phopholipid in the diagram	
	3. explanation of why the phospholipids are orientated the way they are e.g. heads attracted to water OR tails repelled by water ;	3. CCEPT ref to heads being hydrophilic OR tails hydrophobic OR explained in terms of polarity	
	4. proteins in the membrane (described / shown) ;		
	5. idea of two different locations of proteins e.g. extrinsic, intrinsic, transmembrane ;	5. If only one protein located then still get Mp4	
	6. glycoproteins / glycolipids (described / shown) ;		
	7. idea of cholesterol within the membrane (described / shown) ;		(5)

- Use your knowledge of the structure and properties of cell membranes to explain the permeability of non-polar molecules

Question Number	Answer	Mark
3(b)(iv)	1. reference to phospholipid bilayer ; 2. reference to hydrophobic nature (of bilayer / tails) ; 3. idea that {non-polar molecules / molecules that have high solubility in oil compared with water} will pass through the membrane more readily **OR** idea that {polar molecules / molecules with low solubility in oil relative to water} will pass through less readily ; 4. idea that permeability linked to readiness to dissolve ; 5. reference to {fluidity / movement} of phospholipids ;	**max (3)**

- Explain how the properties of molecule A (Phospholipid) contribute to the structure of the cell membrane.

Question Number	Answer	Mark
4(a)(ii)	1. reference to {hydrophilic / polar / charged} part ; 2. reference to {hydrophobic / non polar / uncharged} part ; 3. reference to orientation of molecule in relation to water; 4. idea that aqueous environment is {on two sides / cytoplasm and {environment / tissue fluid / eq}} ;	**max (3)**

- Some proteins in the cell membrane are involved in active transport and facilitated diffusion. Describe the role of proteins in these cell transport mechanisms.

Question Number	Answer	Mark
4(b)	**Active transport:** 1. idea that molecule {binds / fits into} {protein / carriers} ; 2. idea that {protein / carrier} changes shape ; 3. (molecules move) against a concentration gradient / eq ; 4. reference to use of {ATP / energy} ; [Submax 2 marks] **Facilitated diffusion:** 5. reference to proteins as {channels / gates / pores / carriers} ; 6. idea that {channels can open or close / carriers change shape} ; 7. for {large / polar / charged} molecules (to pass through membrane) ; 8. (molecules move) down a concentration gradient / eq ; [Submax 2 marks]	**max (3)**

Question Number	Answer	Mark
4(c)(ii)	1. idea that {phospholipids / molecule A} allow {fluidity / movement/ eq} ; 2. idea that {fluidity / movement / eq} allow membranes to fuse; 3. idea that {fluidity / movement / eq} allows protein to {move / intermingle / eq} ;	**max (2)**

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